

Fractionalized altermagnets: From neighboring and altermagnetic spin liquids to spin-symmetric band splitting

João Augusto Sobral , Subrata Mandal , and Mathias S. Scheurer 
 Institute for Theoretical Physics III, University of Stuttgart, 70550 Stuttgart, Germany



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We study quantum-fluctuation-driven fractionalized phases in the vicinity of altermagnetic order. First, the long-range magnetic orders in the vicinity of collinear altermagnetism are identified; these feature a noncoplanar orbital altermagnet, which has altermagnetic symmetries in spin-rotation-invariant observables. We then describe neighboring fractionalized phases with topological order reached when quantum fluctuations destroy long-range spin order, within Schwinger boson theory and an $SU(2)$ gauge theory of fluctuating magnetism. Discrete symmetries remain broken in some of the fractionalized phases, with the orbital altermagnet becoming an altermagnetic spin liquid. We compute the electronic spectral function in the doped system, which is characterized by split Fermi surfaces with preserved spin-rotation symmetry.

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I. INTRODUCTION

In the last few years, there has been an increasing amount of research on altermagnets (AMs) [1–67]. Unlike antiferromagnets, the total magnetic moment in AMs vanishes not as a result of translations combined with time-reversal Θ symmetry but instead due to the product Θg , where g is a point-group symmetry, e.g., fourfold rotational symmetry $g = C_{4z}$ on the square lattice [2]. While many of the above-mentioned studies deal with the nontrivial impact of altermagnetic moments on quantum physics, e.g., on the electronic band structure or on the properties of superconductivity, studying the impact of quantum fluctuations on the altermagnetic order itself is far less explored. In this work we address the regime of strong quantum fluctuations which restore spin-rotation (SR) invariance in AMs and in other neighboring nontrivial magnetic orders, in both insulators and metallic phases. This is also relevant in light of recent experiments that indicate the interplay between altermagnetism and frustration for thin films of Mn_5Si_3 [68,69] and $CsCr_3Sb$ [48].

More specifically, we start from magnetically ordered phases close to a square-lattice AM in a checkerboard Heisenberg model [42,44,70–78] supplemented by ring-exchange terms. Apart from the collinear AM and other noncollinear states that preserve all symmetries modulo spin rotations, i.e., will appear symmetric in SR-invariant observables, there are also nematic spin orders and an orbital AM. The last phase, which is connected to the collinear AM by a second-order phase transition in our model, is characterized by scalar spin chiralities which are staggered on the lattice in a way that preserves ΘC_{4z} and translation, akin to the spin order of the prototypical AM. We characterize the associated spin-liquid

phases obtained by quantum melting any of these magnetic orders. We present Schwinger boson (SB) [71,79–83] *Ansätze* for each of them and demonstrate that these spin liquids inherit discrete broken symmetries. Most notably, the orbital AM becomes an altermagnetic spin liquid which is characterized by the same altermagnetic spatial arrangement of scalar spin chiralities. Finally, we supplement this with an $SU(2)$ gauge theory [84–91], which allows us to generalize these fractionalized orders to the doped system. Remarkably, the spectral function of the itinerant electrons exhibits peaks similar to the spin splitting in AMs, while maintaining SR invariance.

II. CLASSICAL PHASE DIAGRAM

Although the concepts are more general, for concreteness, we start our discussion with a minimal Heisenberg model. To be able to capture AMs, we need two sublattices A and B per unit cell, which we place on the bonds of a square lattice [see Fig. 1(a)]. Denoting the spin operators on bond i by $\hat{S}_i$, the Hamiltonian reads

$$\begin{aligned} \mathcal{H} = & J_1 \sum_{\langle i,j \rangle} \hat{S}_i \cdot \hat{S}_j + J_2 \sum_{\langle\langle i,j \rangle\rangle} \hat{S}_i \cdot \hat{S}_j \\ & + K \sum_{\substack{j \text{---} k \\ i \text{---} l \\ \text{square}}} [(\hat{S}_i \cdot \hat{S}_j)(\hat{S}_k \cdot \hat{S}_l) + (\hat{S}_i \cdot \hat{S}_l)(\hat{S}_k \cdot \hat{S}_j)]. \end{aligned} \quad (1)$$

If we only included the first term, the nearest-neighbor exchange interaction, which we choose to obey $J_1 > 0$ here, the model would become identical to a square-lattice antiferromagnet (with the lattice rotated by 45°). Adding the next-nearest-neighbor couplings J_2 [blue solid lines in Fig. 1(a)] leads to the checkerboard model [42,44,70–78]. To lift degeneracies and stabilize additional phases, we further include the ring-exchange term in the second line of Eq. (1); it couples any four spins i, j, k , and l on bonds with a common vertex in a way that preserves the C_{4v} point-group symmetries,

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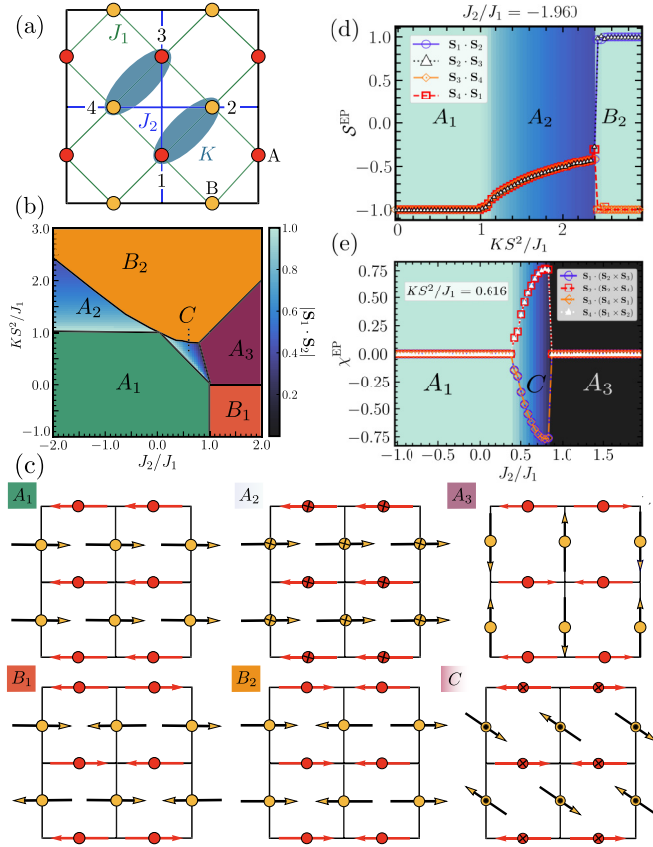


FIG. 1. (a) Bravais lattice (black squares), spins (red and orange sublattices), and couplings J_1 , J_2 , and K of the model (1). (b) Classical phase diagram, with the corresponding phases defined in (c) and Table I. (d) Nearest-neighbor spin products $\mathcal{S}^{\text{EP}}$ and (e) scalar spin chiralities χ^{EP} along two different one-dimensional cuts through (b). The background coloring in (d) and (e) refers to $|\mathbf{S}_1 \cdot \mathbf{S}_2|$ as defined in the color bar in (b).

time-reversal Θ symmetry, SR symmetry, and Bravais-lattice translational symmetry.

We first analyze the magnetic phases in the classical limit and make the *Ansatz*

$$\mathcal{S}_\ell(\mathbf{R}) \propto \hat{\mathbf{a}} \cos(\mathbf{Q} \cdot \mathbf{R} - \varphi_\ell) + \hat{\mathbf{b}} \sin(\mathbf{Q} \cdot \mathbf{R} - \varphi_\ell) + \hat{\mathbf{c}} \eta_\ell \quad (2)$$

for the direction of the magnetization on sublattice $\ell = A, B$ in the square-lattice unit cell $\mathbf{R}$; here $\hat{\mathbf{a}}$, $\hat{\mathbf{b}}$, and $\hat{\mathbf{c}}$ are three arbitrary three-component orthonormal vectors. Without loss of generality, we set $\varphi_A = 0$ and $\varphi_B \equiv \varphi$ and minimize the classical energy with respect to the variational parameters $\{Q_x, Q_y, \varphi, \eta_A, \eta_B\}$ (see Appendix A).

In agreement with the nearest-neighbor square-lattice antiferromagnet model, the system favors an antiparallel alignment of the spins on the two sublattices, $\mathcal{S}_A(\mathbf{R}) = -\mathcal{S}_B(\mathbf{R}) = \hat{\mathbf{a}}$, for small K and J_2 [see phase A_1 in Figs. 1(b) and 1(c)]. Importantly though, finite K and J_2 in Eq. (1) break the artificial translational symmetry at $J_2 = K = 0$ relating the two sublattices A and B ; instead, they are related by a C_{4z} rotational symmetry in the full model, making A_1 a collinear AM. It has vanishing total magnetization $\mathbf{M} = \sum_{\mathbf{R}, \ell} \mathcal{S}_\ell(\mathbf{R}) = 0$, as guaranteed by $C_{4z}\Theta$. Modulo global spin rotations, all magnetic point-group symmetries are pre-

served, which will also play an important role in our later discussion of nearby spin liquids. Since this implies that all SR-invariant observables must also respect all symmetries of our model, we can conveniently probe these symmetries using the nearest-neighbor scalar and (counterclockwise) triple products $\mathcal{S}^{\text{EP}} = (\mathbf{S}_1 \cdot \mathbf{S}_2, \mathbf{S}_2 \cdot \mathbf{S}_3, \mathbf{S}_3 \cdot \mathbf{S}_4, \mathbf{S}_4 \cdot \mathbf{S}_1)^T$ and $\chi^{\text{EP}} = (\mathbf{S}_1 \cdot (\mathbf{S}_2 \times \mathbf{S}_3), \mathbf{S}_2 \cdot (\mathbf{S}_3 \times \mathbf{S}_4), \mathbf{S}_3 \cdot (\mathbf{S}_4 \times \mathbf{S}_1), \mathbf{S}_4 \cdot (\mathbf{S}_1 \times \mathbf{S}_2))^T$, respectively, where the sites are labeled as in Fig. 1(a). Indeed, we can see from Table I that all components of $\mathcal{S}^{\text{EP}}$ are identical (isotropic and non-nematic) and $\chi^{\text{EP}} = 0$ (preserved Θ and nonchiral).

Turning on sufficiently large $K > 0$, it becomes energetically favorable to reduce the magnitude of the nearest-neighbor spin products, by developing a finite canting. This leads to the second-order phase transition into the canted AM A_2 , $\mathcal{S}_{A,B}(\mathbf{R}) = \pm \hat{\mathbf{a}} + \eta \hat{\mathbf{c}}$ [see Figs. 1(b)–1(d)]. As summarized in Table I, this leads to a finite magnetization $\mathbf{M} \neq 0$ while the symmetries (modulo spin rotation) are identical to those of A_1 . In the phase diagram in Fig. 1(b), there is one more fully symmetric phase, denoted by A_3 ; the sufficiently large J_2 leads to finite $Q_{x,y} = \pi$ while positive K favors perpendicular spins in the two sublattices ($\varphi = \pi/2$) [see Fig. 1(c)]. As a result of translational symmetry, the magnetization vanishes, and the state can be thought of as an antiferromagnet with four spins in the enlarged unit cell.

Furthermore, the phase diagram contains two nematic phases, denoted by $B_{1,2}$, which break C_{4z} rotational symmetry in SR-invariant observables while retaining Θ symmetry [see Fig. 1(c) and Table I]. The phase B_1 is reached via a first-order transition from A_3 by reversing the sign of K , leading to parallel spins in the unit cell ($\varphi = 0$). Meanwhile B_2 is stabilized when $K > 0$ dominates the energetics, leading to a phase with only one of Q_x or Q_y nonzero (and equal to π).

Finally, in the most frustrated region of the phase diagram in Fig. 1(b), a noncoplanar phase C is stabilized. As can be seen in Fig. 1(e), this state is reached from the collinear AM A_1 via a second-order phase transition, where the scalar spin chiralities become nonzero, breaking Θ in SR-invariant observables. Importantly, the chiralities are staggered, $\chi^{\text{EP}} \propto (-1, 1, -1, 1)$, in a way that preserves $C_{4z}\Theta$ (and $\sigma_v\Theta$). As such, phase C behaves like an AM in SR-invariant observables, i.e., is associated with (when adding charge carriers; see Sec. IV below) a special arrangement of orbital currents [89,91–93], where the corresponding orbital magnetic moments are finite locally but vanish globally, not as a result of translational symmetry but due to $C_{4z}\Theta$. For this reason, we refer to phase C as an orbital AM [38].

III. NEARBY SPIN LIQUIDS

Having established the classical phase diagram, we next discuss spin-liquid phases that can be reached from the respective magnetically ordered phases once quantum fluctuations restore SR invariance. To this end, we will first use a SB description [79,80] where the spin operators in Eq. (1) are rewritten as $\hat{\mathbf{S}}_i = \hat{b}_{i\sigma}^\dagger \boldsymbol{\sigma}_{\sigma\sigma} \hat{b}_{i\sigma} / 2$, introducing a local $U(1)$ gauge redundancy $\hat{b}_{j,\sigma} \rightarrow e^{i\phi_j} \hat{b}_{j,\sigma}$; here $\boldsymbol{\sigma}$ are Pauli matrices and $\hat{b}_{i,\sigma}$ canonical bosons. Since this enlarges the Hilbert space of spin- S operators, the local constraint $\hat{n}_i = 2S$ and $\hat{n}_i = \hat{b}_{i\sigma}^\dagger \hat{b}_{i\sigma}$ is imposed. Decoupling the resulting boson-boson

TABLE I. Summary of the magnetic phases for the spin model (1). We list one possible set of parameters for Eq. (2) and the values of $\mathcal{S}^{\text{EP}}$ and χ^{EP} ($s_0, s'_0, \xi_0 \in \mathbb{R}^+$ are nonuniversal parameters depending on the couplings). We further indicate whether the total magnetization $\mathbf{M}$ is finite and the magnetic point symmetries of spin-rotation-invariant observables (through their generators); these are also the symmetries of the associated spin liquids in Fig. 2 and the itinerant fractionalized phases with topological order discussed below. The invariant gauge group (IGG) of these two descriptions coincide and are listed in the last column.

Label	Type	$\mathcal{S}^{\text{EP}}$	χ^{EP}	$(Q_x, Q_y, \varphi, \eta_A, \eta_B)$	$\mathbf{M}$	Symmetries	IGG
A_1	collinear AM (isotropic, nonchiral)	$-(1, 1, 1, 1)$	$(0,0,0,0)$	$(0, 0, \pi, 0, 0)$	0	C_{4z}, σ_v, Θ	U(1)
A_2	canted AM (isotropic, nonchiral)	$-(s_0, s_0, s_0, s_0)$	$(0,0,0,0)$	$(0, 0, \pi, \eta, \eta)$	$\neq 0$	C_{4z}, σ_v, Θ	$\mathbb{Z}_2$
A_3	orthogonal AFM (isotropic, nonchiral)	$(0, 0, 0, 0)$	$(0,0,0,0)$	$(\pi, \pi, \pi/2, 0, 0)$	0	C_{4z}, σ_v, Θ	$\mathbb{Z}_2$
B_1	collinear nematic (nonchiral)	$(-1, 1, -1, 1)$	$(0,0,0,0)$	$(\pi, \pi, 0, 0, 0)$	0	C_{2z}, σ_d	U(1)
B_2	collinear odd parity (nonchiral)	$(1, 1, -1, -1)$	$(0,0,0,0)$	$(\pi, 0, 0, 0, 0)$	0	σ_v, Θ	U(1)
C	orbital AM (isotropic, staggered chirality)	$-(s'_0, s'_0, s'_0, s'_0)$	$(-\xi_0, \xi_0, -\xi_0, \xi_0)$	$(\pi, \pi, \pi/2, \eta, -\eta)$	0	$C_{4z}\Theta, \sigma_v\Theta$	$\mathbb{Z}_2$

interactions in the spin Hamiltonian leads to a free boson Hamiltonian

$$\mathcal{H}_b = \frac{1}{2} \sum_{i,j} (\mathcal{B}_{ij}^* \hat{b}_{i\sigma} \hat{b}_{j\sigma}^\dagger - \mathcal{A}_{ij}^* \sigma \hat{b}_{i\sigma} \hat{b}_{j-\sigma} + \text{H.c.}) - \mu \sum_i \hat{n}_i,$$

where $\mathcal{A}_{ij}, \mathcal{B}_{ij} \in \mathbb{C}$ can be computed within self-consistent mean-field theory; the same applies for the Lagrange multiplier μ introduced to treat the local constraint on average. However, as this mean-field approach is typically not quantitatively reliable, here we take a different perspective: We systematically study possible *Ansätze* for $\mathcal{A}_{ij}$ and $\mathcal{B}_{ij}$, starting with only nearest-neighbor terms and adding further-neighbor couplings until we find a spin liquid where Bose condensation (see Appendix B) leads to a (apart from global spin rotations) unique magnetic ground state that is identical to one of the phases in Fig. 1(b) and Table I; in this way, we can associate a spin liquid with each of these magnetic phases.

For all of them, restricting the model to nearest- and next-nearest-neighbor bonds i and j in $\mathcal{A}_{ij}$ and $\mathcal{B}_{ij}$ which (upon proper gauge choice) do not increase the unit cell turns out to be sufficient. We define all of these *Ansätze* diagrammatically in Fig. 2. For instance, we see that the *Ansatz* for A_1 only involves $\mathcal{A}_{ij}$ on nearest-neighbor bonds. Being left invariant by a U(1) gauge transformation consisting of phases

of alternating signs in the two sublattices, it is a U(1) spin liquid. It is straightforward to show that the *Ansatz* is invariant (modulo gauge transformations) under all symmetries of the model (1), which we also demonstrated by computing the expectation values of $\mathcal{S}^{\text{EP}}$ and χ^{EP} in the spin-liquid phase: Indeed, as also indicated in Fig. 2, we get $\langle \mathcal{S}_l^{\text{EP}} \rangle = \text{const}$ and $\langle \chi^{\text{EP}} \rangle = 0$. The *Ansatz* for A_2 has additional imaginary $\mathcal{B}_{ij}$ on nearest-neighbor bonds, which not only control the canting in the associated magnetically ordered phase but also lead to a $\mathbb{Z}_2$ spin liquid. Just like the *Ansatz* for A_1 , this $\mathbb{Z}_2$ spin liquid also respects all symmetries of the Hamiltonian. Also for the third magnetic phase A_3 , which respects all symmetries modulo spin rotations, the associated SB *Ansatz*, now with second-nearest-neighbor terms (see Fig. 2), leads to a fully symmetric $\mathbb{Z}_2$ spin liquid.

In contrast, the spin liquids associated with the remaining phases have reduced symmetries. As one would expect, the spin liquids of the nematic magnetic phases $B_{1,2}$ are nematic themselves: As can be seen from the correlators $\langle \mathcal{S}^{\text{EP}} \rangle$ shown along with the *Ansätze* in Fig. 2 and upon noting that still $\langle \chi^{\text{EP}} \rangle = 0$ (Θ is not broken), they obey the exact same symmetries as listed in Table I for $B_{1,2}$. Finally, the spin liquid associated with the orbital AM also has reduced symmetries. We find isotropic correlators $\langle \mathcal{S}_l^{\text{EP}} \rangle = \text{const}$ but staggered scalar spin chiralities $\langle \chi^{\text{EP}} \rangle \propto (-1, 1, -1, 1)$; while SR invariance is restored, the altermagnetic arrangement of orbital moments (being translation invariant but odd under C_{4z}) survives in the neighboring spin-liquid phase, which can thus be thought of as an altermagnetic spin liquid. This phase hosts vison excitations associated with local distortions of the magnetization pattern [94] that could be detected with local experimental probes [95].

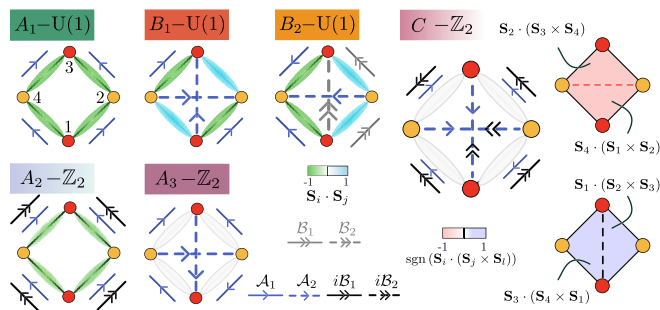


FIG. 2. Diagrammatic definition of the Schwinger boson *Ansätze*, where blue (black and gray) arrows indicate real $\mathcal{A}_{ij}$ (imaginary and real $\mathcal{B}_{ij}$) on the respective bonds, for the spin liquids associated with the magnetic phases in Figs. 1(b) and 1(c). We further indicate the expectation values of $\mathcal{S}^{\text{EP}}$ using colored ellipses and, for phase C , of χ^{EP} (only showing two triple products in each of the nearest two unit cells for clarity) in the respective spin liquids. The patterns extend through the entire lattice.

IV. FRACTIONALIZED ITINERANT SYSTEMS

One of the central aspects of AMs is their impact on the spectrum of itinerant electrons, creating a momentum-dependent spin splitting of their bands. We address here the related question but in the regime where the spin degrees of freedom are in a spin-liquid state. To this end, we assume that mobile electrons are added to the model studied above, which we describe by the creation operators $\hat{c}_{\mathbf{R},\ell,\sigma}^\dagger$ of spin- σ electrons on the bond $(\mathbf{R}, \ell)$ in Fig. 1(a). We include nearest- (t_1) and next-nearest-neighbor (t_2) hopping [19] and couple

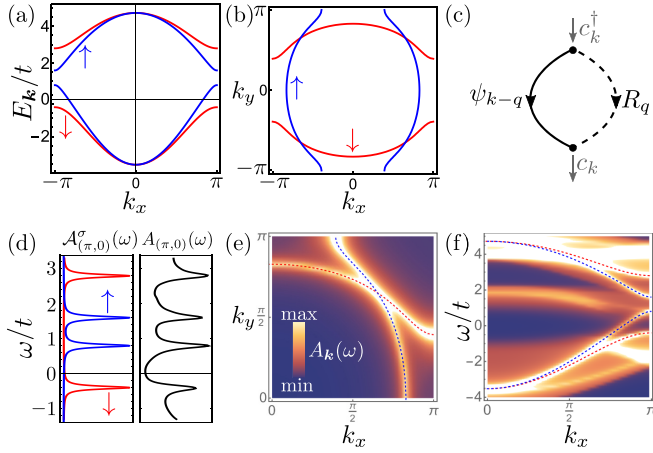


FIG. 3. (a) Band structure along the $k_y = 0$ line and (b) Fermi surfaces for the long-range A_1 order. (c) The electronic Green's function is a convolution of the chargin and spinon Green's function. (d) Electronic spectral function in the presence of long-range order [$A_k^\sigma(\omega)$, left] and in the fractionalized phase with spin-rotation invariance and topological order [$A_k(\omega)$, right] at $\mathbf{k} = (\pi, 0)$. The spectral function in the fractionalized phase is shown at $\omega = 0$ as a function of (e) momentum $\mathbf{k}$ and (f) energy ω and k_x at $k_y = 0$. We use $t_2/t_1 = 0.3$, chemical potential $\mu/t_1 = -1.2$, $H_0/t_1 = 1$, temperature $T/t_1 = 0.1$, and a lifetime broadening of $\eta/t_1 = 0.05$.

the electrons locally to the magnetic moments $\Phi_{\mathbf{R},\ell}$ as

$$\mathcal{S}_c = \int d\tau \sum_{\mathbf{R},\ell} c_{\mathbf{R},\ell,\sigma}^\dagger \sigma_{\sigma,\sigma'} c_{\mathbf{R},\ell,\sigma'} \cdot \Phi_{\mathbf{R},\ell}, \quad (3)$$

where we switched to a path-integral description with imaginary time τ (see Appendix C for more details). Assuming nonfluctuating long-range order $\Phi_{\mathbf{R},\ell} \rightarrow S_\ell(\mathbf{R})$ and taking, e.g., the collinear AM order $S_{A,B}(\mathbf{R}) = \pm \hat{e}_z$ of phase A_1 , we obtain the bands and Fermi surface with the characteristic spin splitting of AMs shown in Figs. 3(a) and 3(b).

To describe the regime of strong quantum fluctuations, where SR invariance is restored, we employ the approach of Refs. [84–91], where a transformation into a rotating reference frame in spin space is performed using the dynamical bosonic SU(2) matrices $R(\tau)$ in $c_{\mathbf{R},\ell,\sigma}(\tau) = [R_{\mathbf{R},\ell}(\tau)]_{\sigma,\alpha} \psi_{\mathbf{R},\ell,\alpha}(\tau)$. Physically, the R capture the spin degrees of freedom and are directly related to the bosons in our SB approach above, while the charge is carried by the fermionic chargin ψ . This rewriting introduces a local SU(2) redundancy as the physical degrees of freedom are unchanged under $R_{\mathbf{R},\ell} \rightarrow R_{\mathbf{R},\ell} V_{\mathbf{R},\ell}^\dagger$ and $\psi_{\mathbf{R},\ell,\alpha} \rightarrow V_{\mathbf{R},\ell} \psi_{\mathbf{R},\ell}$ for arbitrary SU(2) matrices $V_{\mathbf{R},\ell}$. We introduce the field $\mathbf{H}_{\mathbf{R},\ell}$ via

$$\mathbf{H}_{\mathbf{R},\ell} \cdot \sigma = \Phi_{\mathbf{R},\ell} \cdot R_{\mathbf{R},\ell}^\dagger \sigma R_{\mathbf{R},\ell}, \quad (4)$$

which can be intuitively thought of as Φ represented in a rotating reference frame; in the emergent gauge theory, $\mathbf{H}_{\mathbf{R},\ell}$ plays the role of the Higgs field. By construction, Eq. (3) can be compactly restated as a coupling between the chargin and the Higgs field [see Eq. (C6)].

If both $R_{\mathbf{R},\ell}$ and $\mathbf{H}_{\mathbf{R},\ell}$ condense, we will just reobtain long-range magnetic order; upon choosing a gauge where $\langle R_{\mathbf{R},\ell} \rangle = \mathbb{1}$, the magnetic texture is equal to the texture of

$\langle \mathbf{H}_{\mathbf{R},\ell} \rangle$, allowing us to connect the Higgs-field configurations to the classical magnetic orders studied above. When the bosons are gapped, $\langle R_{\mathbf{R},\ell} \rangle = 0$, SR invariance is restored, even when the Higgs field is still condensed ($\langle \mathbf{H}_{\mathbf{R},\ell} \rangle \neq 0$), which defines a phase with topological order. Despite the preserved SR symmetry, discrete symmetries can still be broken [89,91]. As the symmetry analysis is mathematically identical to studying whether the associated classical magnetic orders preserve a symmetry modulo spin rotations, we conclude that the associated doped spin liquids have the same symmetries as those listed in Table I. More specifically, in agreement with our Schwinger boson study, the fractionalized phases associated with $A_{1,2,3}$ will respect to all symmetries, those with $B_{1,2}$ will be nematic and the one for phase C will exhibit altermagnetic loop currents [89,91]. Notably, both approaches also yield the same invariant gauge groups.

The gauge theory further allows one to compute the electronic spectral function in those phases [90,96], which is obtained as a convolution of the chargin and spinon Green's functions [see Fig. 3(c)]. Focusing for concreteness on the fractionalized phase of A_1 , we compare the resulting spectral function with the one in the magnetically ordered phase in Fig. 3(d). As a result of the restored SU(2) SR symmetry, the spectral function $A_k(\omega)$ in the fractionalized phase does not depend on spin and yet has peaks centered around the same frequencies as the state with long-range AM order. This signals fractionalization since, for a given spin species, each peak in the right panel is associated with half an electron. When plotting $A_k(\omega)$ as a function of $\mathbf{k}$ at the Fermi level, i.e., $\omega = 0$, in Fig. 3(e) we clearly see the characteristic $\mathbf{k}$ -dependent splitting of the Fermi surfaces. The band splitting persists at energies away from the Fermi level [see Fig. 3(f)], which further reveals that there are additional (incoherent) high-energy features associated with the presence of spinons. A combination of quantum oscillations (probing the chargin) and (spin-resolved) photoemission [60] [measuring $A_k(\omega)$] can identify signatures of such a fractionalized state experimentally.

V. CONCLUSION

In summary, our work shows that driving altermagnets into the frustrated regime with competing couplings in scenarios where quantum fluctuations play an important role gives rise to very rich physics, such as orbital altermagnets, various interesting spin liquids, including states with altermagnetic correlations in the charge sector, as well as the emergence of spin-symmetric band splitting. Given the increasingly large number of candidate materials for altermagnets [26], which typically exhibit multiple complex moments in their unit cell, it seems plausible that altermagnetic systems with a sufficient degree of frustration [48,68,69] can be identified.

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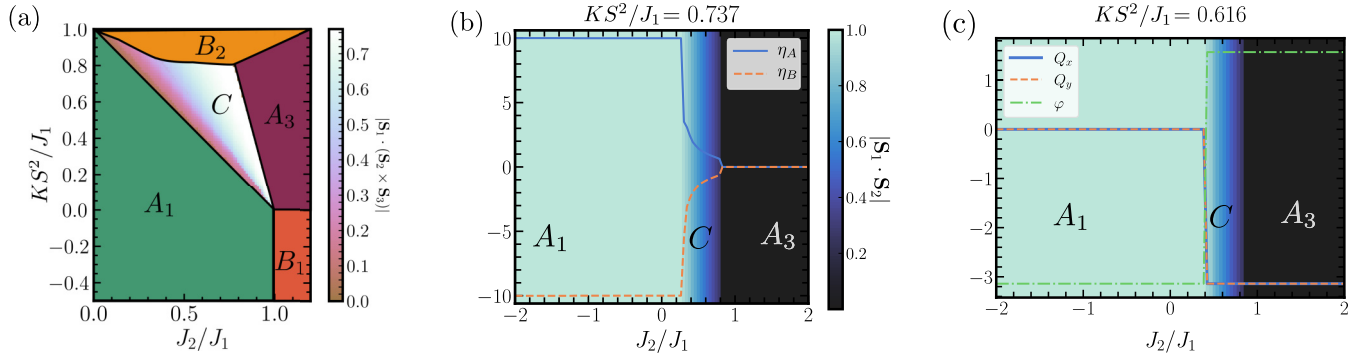


FIG. 4. Phase diagram of the model (1) in the classical limit $S \rightarrow \infty$, focused around the region where the orbital AM C is located. (b) and (c) One-dimensional cuts for the phase transition $A_1 \rightarrow C \rightarrow A_3$ from the perspective of the variational parameters α_{conv} . The colormap in (a) refers to the absolute value of the scalar spin chirality $|\mathbf{S}_1 \cdot (\mathbf{S}_2 \times \mathbf{S}_3)|$, whereas the background color in (b) and (c) refers to the absolute value of the nearest-neighbor spin product $|\mathbf{S}_1 \cdot \mathbf{S}_2|$.

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APPENDIX A: CLASSICAL PHASE DIAGRAM

As discussed in the main text, to study the ordered phases for $\mathcal{H}$ defined in Eq. (1), we start from the classical limit $S \rightarrow \infty$. For this, we consider the classical *Ansatz* (2) for the two sublattices in the unit cell given by

$$\begin{aligned} \mathbf{S}_{A,\alpha} &= (\cos(\mathbf{Q} \cdot \mathbf{R}), \sin(\mathbf{Q} \cdot \mathbf{R}), \eta_A)^T / \sqrt{1 + \eta_A^2}, \\ \mathbf{S}_{B,\alpha} &= (\cos(\mathbf{Q} \cdot \mathbf{R} - \varphi), \sin(\mathbf{Q} \cdot \mathbf{R} - \varphi), \eta_B)^T / \sqrt{1 + \eta_B^2}, \end{aligned} \quad (\text{A1})$$

where $\alpha = \{Q_x, Q_y, \varphi, \eta_A, \eta_B\}$ is a set of variational parameters yet to be determined. We determine the classical phase diagram by minimizing the classical energy $\mathcal{H}(\mathbf{S}_{A,\alpha} \cdot \mathbf{S}_{B,\alpha})$ expression with respect to the parameters α , using a combination of grid search over the variational parameters' space plus gradient-descent-like optimization methods to find energetically favored configurations.

The first step for the grid search method consists of defining an energy grid with a total of L_{GD} equally spaced values for each of the variational parameters in α , with the domains for the initial values α_0 defined as $Q_x^0, Q_y^0, \varphi^0 \in [-\pi, \pi]$ and $\eta_A^0, \eta_B^0 \in [-10, 10]$. For each pair of coupling constants $(J_2/J_1, KS^2/J_1)$, we identify the configurations α_0 which have the lowest energy within the computed energy grid. These configurations are then selected as a starting point for the gradient-descent algorithm. These sets of parameters α_0 are iteratively updated with the standard formula $\alpha = \alpha_0 - \lambda \partial \mathcal{H} / \partial \alpha$. Here λ is the learning rate, and each iteration constitutes one epoch. Our simulations included $\lambda \in [0.1, 0.001]$ over different runs up to 4×10^4 epochs. We repeated this process for varying energy grid sizes $L \in [10, 40]$. Additionally, we also considered one variation of the standard gradient-descent method called Adam, which introduces adaptable

learning rates for each of the variational parameters [97]. After convergence, the minimum-energy configurations at each $(KS^2/J_1, J_2/J_1)$ point are characterized by the sets α_{conv} . Their energies were then compared to determine the phase boundaries in Fig. 1(b). A particular set for each of the phases in the classical phase diagram in Fig. 1(b) can be seen in Table I, as well as their spin textures $\mathbf{S}_{A,B}(\alpha_{\text{conv}})$ in Fig. 1(c). As stated in the main text, each magnetic phase can also be characterized by two sets of observables: the nearest-neighbor spin products on the elementary plaquette given by $\mathcal{S}^{\text{EP}}$ and the scalar spin chiralities χ^{EP} . The latter are crucial for characterizing non-coplanar textures [89,98,99]. The orbital AM C is the only phase in the classical phase diagram with nonzero scalar spin chirality. It has a staggered pattern χ^{EP} (see Table I) that evolves continuously for varying $(KS^2/J_1, J_2/J_1)$, as can be seen in Figs. 1(e) and 4(a). The continuous change of this observable is accompanied by a similar behavior from the perspective of the variational parameters η_A and η_B obtained in α_{conv} , which always obey the constraint $\eta_A + \eta_B = 0$ [see Fig. 4(b)]. Likewise, the nearest-neighbor spin products $\mathcal{S}^{\text{EP}}$ also vary continuously in C and A_2 [see Figs. 1(b) and 1(d) and the background color in Figs. 4(b) and 4(c)]. The other phases A_1, B_1, B_2 , and A_3 have fixed values for $\mathcal{S}^{\text{EP}}$ and zero scalar spin chirality (see Table I).

APPENDIX B: SCHWINGER BOSON ANSÄTZE

1. General formalism

In this Appendix we discuss in more detail the SB description of the spin-liquid states considered in the main text, as well as their associated magnetic orders obtained after the bosonic condensation. We start by rewriting the Hamiltonian (1), without the ring-exchange term K , in terms of the operators

$$\hat{S}_i = \frac{1}{2} \sum_{\sigma, \sigma' = \uparrow, \downarrow} \hat{b}_{i\sigma}^\dagger \sigma_{\sigma\sigma'} \hat{b}_{i\sigma'}. \quad (\text{B1})$$

The inclusion of only J_1 and J_2 exchange interactions in the SB formalism turns out to be sufficient to identify *Ansätze* corresponding to all the magnetic phases present in the classical phase diagram in Fig. 1(b). Using the SU(2) completeness

relation

$$\sigma_{\mu\mu'} \cdot \sigma_{\nu\nu'} = \sum_{k=1}^3 \sigma_{\mu\mu'}^k \sigma_{\nu\nu'}^k = 2\delta_{\mu\nu'}\delta_{\mu'\nu} - \delta_{\mu\mu'}\delta_{\nu\nu'}, \quad (\text{B2})$$

we can rewrite the dot product of spins as $\hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j = \hat{\mathcal{B}}_{ij}^\dagger \mathcal{B}_{ij} - \hat{\mathcal{A}}_{ij}^\dagger \mathcal{A}_{ij}$, where $\hat{\mathcal{A}}_{ij}$ and $\hat{\mathcal{B}}_{ij}$ are bond operators defined as

$$\hat{\mathcal{A}}_{ij} = \frac{1}{2} \sum_{\sigma, \sigma'=\uparrow, \downarrow} \hat{b}_{i\sigma}(i\sigma_y)_{\sigma\sigma'} \hat{b}_{j\sigma'}, \quad \hat{\mathcal{B}}_{ij} = \frac{1}{2} \sum_{\sigma=\uparrow, \downarrow} \hat{b}_{i\sigma} \hat{b}_{j\sigma}^\dagger. \quad (\text{B3})$$

These are typically referred to as pairing and hopping operators, respectively. After a Hubbard-Stratonovich transformation, we obtain the decoupled mean-field Hamiltonian

$$\mathcal{H}_{\text{MF}} = \sum_{i,j,\sigma} \frac{1}{2} [(\mathcal{B}_{ij}^* \hat{b}_{i\sigma} \hat{b}_{j\sigma}^\dagger - \mathcal{A}_{ij}^* \sigma \hat{b}_{i\sigma} \hat{b}_{j-\sigma} + \text{H.c.}) + (|\mathcal{A}_{ij}|^2 - |\mathcal{B}_{ij}|^2)/J_{ij}] - \mu \sum_i \hat{b}_{i\sigma}^\dagger \hat{b}_{i\sigma}, \quad (\text{B4})$$

where μ is the chemical potential and $\mathcal{A}_{ij}$ and $\mathcal{B}_{ij}$ are the mean-field values of the bond operators, i.e.,

$$\langle \hat{\mathcal{A}}_{ij} \rangle = \mathcal{A}_{ij}/J_{ij}, \quad \langle \hat{\mathcal{B}}_{ij} \rangle = \mathcal{B}_{ij}/J_{ij}. \quad (\text{B5})$$

Equation (B4) has the same functional form as the Hamiltonian $\mathcal{H}_b$ mentioned in the main text, apart from the constant energy shift $\mathcal{H}_{\text{MF}}^0 = \sum_{i,j,\sigma} (|\mathcal{A}_{ij}|^2 - |\mathcal{B}_{ij}|^2)/J_{ij}$. The mean-field parameters $\mathcal{A}_{ij}$ and $\mathcal{B}_{ij}$ can assume complex values and satisfy the constraints $\mathcal{A}_{ij} = -\mathcal{A}_{ji}$ and $\mathcal{B}_{ij} = \mathcal{B}_{ji}^*$. The Hamiltonian (B4) exhibits an emergent U(1) gauge symme-

try, which can be clearly seen by noticing that $\mathcal{H}_{\text{MF}}$ remains invariant under a local U(1) gauge transformation defined by

$$\begin{aligned} \hat{b}_{i\sigma} &\rightarrow e^{i\phi_i} \hat{b}_{i\sigma}, \quad \sigma = \uparrow, \downarrow \\ \mathcal{A}_{ij} &\rightarrow e^{i(\phi_i + \phi_j)} \mathcal{A}_{ij}, \\ \mathcal{B}_{ij} &\rightarrow e^{i(\phi_i - \phi_j)} \mathcal{B}_{ij}. \end{aligned} \quad (\text{B6})$$

Consequently, each spin-liquid *Ansatz* can be identified through gauge-invariant fluxes defined as

$$\begin{aligned} \Phi_{ijkl}^{\square AA} &= \arg(\mathcal{A}_{ij} \mathcal{A}_{jk}^* \mathcal{A}_{kl} \mathcal{A}_{li}^*), \\ \Phi_{ijkl}^{\square AB} &= \arg(\mathcal{A}_{ij} \mathcal{A}_{jk}^* \mathcal{B}_{kl} \mathcal{B}_{li}), \\ \tilde{\Phi}_{ijkl}^{\square AB} &= \arg(\mathcal{A}_{ij} \mathcal{B}_{jk}^* \mathcal{A}_{kl}^* \mathcal{B}_{li}), \\ \Phi_{ijk}^\Delta &= \arg(\mathcal{A}_{ij} \mathcal{A}_{jk}^* \mathcal{B}_{ki}), \\ \Phi_{ijk}^{\Delta_2} &= \arg(\mathcal{B}_{ij} \mathcal{A}_{jk} \mathcal{A}_{ki}^*). \end{aligned} \quad (\text{B7})$$

We will henceforth refer to each *Ansatz* according to the notation $[\Phi_{1234}^{\square AA}, \Phi_{1234}^{\square AB}, \tilde{\Phi}_{1234}^{\square AB}, \{\Phi_{123}^\Delta, \Phi_{234}^\Delta, \Phi_{341}^\Delta, \Phi_{412}^\Delta\}, \{\Phi_{123}^{\Delta_2}, \Phi_{234}^{\Delta_2}, \Phi_{341}^{\Delta_2}, \Phi_{412}^{\Delta_2}\}]$.

Focusing on *Ansätze* and gauges where the $\mathcal{A}_{ij}$ and $\mathcal{B}_{ij}$ respect the Bravais-lattice translational symmetry, the Hamiltonian (B4) can be rewritten in momentum space after taking the Fourier transformation $\hat{b}_{r,\alpha,\sigma} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}} \hat{b}_{\mathbf{k},\alpha,\sigma}$, as $\mathcal{H}_{\text{MF}} = \sum_{\mathbf{k}} \Psi_{\mathbf{k}}^\dagger M_{\mathbf{k}}(\mathcal{A}_{ij}, \mathcal{B}_{ij}) \Psi_{\mathbf{k}} + \text{const.}$ Here $\Psi_{\mathbf{k}} = (\hat{b}_{\mathbf{k}A\uparrow}, \hat{b}_{\mathbf{k}B\uparrow}, \hat{b}_{-\mathbf{k}A\downarrow}^\dagger, \hat{b}_{-\mathbf{k}B\downarrow}^\dagger)^T$ is the Nambu spinor and the 4×4 matrix $M_{\mathbf{k}}$ is given by

$$M_{\mathbf{k}}(\mathcal{A}_{ij}, \mathcal{B}_{ij}) = \frac{1}{2} \begin{pmatrix} \mathcal{B}_{24}^* e^{-ik_x} + \mathcal{B}_{24} e^{ik_x} - 2\mu & \tilde{\xi}_{\mathbf{k}}^* & -2i\mathcal{A}_{24} \sin(k_x) & -\xi_{-\mathbf{k}} \\ \tilde{\xi}_{\mathbf{k}} & \mathcal{B}_{13}^* e^{ik_y} + \mathcal{B}_{13} e^{-ik_y} - 2\mu & \xi_{\mathbf{k}} & 2i\mathcal{A}_{13} \sin(k_y) \\ 2i\mathcal{A}_{24}^* \sin(k_x) & \xi_{\mathbf{k}}^* & \mathcal{B}_{24}^* e^{ik_x} + \mathcal{B}_{24} e^{-ik_x} - 2\mu & \tilde{\xi}_{-\mathbf{k}} \\ -\xi_{-\mathbf{k}}^* & -2i\mathcal{A}_{13}^* \sin(k_y) & \tilde{\xi}_{-\mathbf{k}} & \mathcal{B}_{13}^* e^{-ik_y} + \mathcal{B}_{13} e^{ik_y} - 2\mu \end{pmatrix}, \quad (\text{B8})$$

with $\tilde{\xi}_{\mathbf{k}} = \mathcal{B}_{23}^* + \mathcal{B}_{12} e^{-ik_y} + \mathcal{B}_{34} e^{ik_x} + \mathcal{B}_{14} e^{-i(k_y - k_x)}$ and $\xi_{\mathbf{k}} = -\mathcal{A}_{12} e^{-ik_y} + \mathcal{A}_{23} - \mathcal{A}_{34} e^{ik_x} - \mathcal{A}_{14} e^{-i(k_y - k_x)}$.

a. Diagonalization of the quadratic bosonic Hamiltonians

In order to diagonalize $\mathcal{H}_{\text{MF}}$, we consider the Bogoliubov-Valatin transformation, a well-known method for diagonalizing a general quadratic bosonic Hamiltonian [100–103]. For simplicity, we suppress the momentum index $\mathbf{k}$ for now and consider a Hamiltonian H of the form

$$H = \frac{1}{2} \Psi^\dagger M_{\mathbf{k}} \Psi \quad \text{with } \Psi^\dagger = (\hat{b}_1^\dagger, \dots, \hat{b}_N^\dagger, \hat{b}_1, \dots, \hat{b}_N)^T, \quad (\text{B9})$$

with N different values of internal quantum numbers describing degrees of freedom such as spin or sublattice. We then reformulate the bosonic Hamiltonian H in terms of a new set of creation and annihilation operators $\hat{\gamma}_m^\dagger$ and $\hat{\gamma}_m$, respectively. The diagonalizability of the original bosonic Hamiltonian H ensures the existence of such a transformation matrix T that

turns $M_{\mathbf{k}}$ into a simplified diagonal-matrix form in

$$\begin{aligned} H &= \frac{1}{2} \Gamma^\dagger T^\dagger M_{\mathbf{k}} T \Gamma \quad \text{with } T^\dagger M_{\mathbf{k}} T \\ &= \begin{pmatrix} \omega_1 & 0 & \cdots & 0 \\ 0 & \omega_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \omega_{2N} \end{pmatrix}, \quad \Psi = T \Gamma, \\ \hat{\Gamma}^\dagger &\equiv (\hat{\gamma}_1^\dagger, \dots, \hat{\gamma}_N^\dagger, \hat{\gamma}_1, \dots, \hat{\gamma}_N)^T. \end{aligned} \quad (\text{B10})$$

The transformation matrix T can be constructed from the eigenvectors $V(\omega_i)$ of $K_{\mathbf{k}} = \rho_3 M_{\mathbf{k}}$, known as the dynamic matrix. The eigenvalues of the dynamic matrix $K_{\mathbf{k}}$ appear in pairs and are real if $K_{\mathbf{k}}$ is diagonalizable. Furthermore, the pair of eigenvalues is generally related to two eigenvectors, one with a positive norm and the other with a negative

norm, i.e.,

$$V^\dagger(\omega_i)\rho_3V(\omega_i) = 1, \quad V^\dagger(-\omega_i)\rho_3V(-\omega_i) = -1, \\ \text{where } \rho_3 = \begin{pmatrix} \mathbb{1}_{N \times N} & 0 \\ 0 & -\mathbb{1}_{N \times N} \end{pmatrix}. \quad (\text{B11})$$

Finally, bosonic commutation relations constrain the transformation matrix T to be paraunitary, i.e., $T\rho_3T^\dagger = \rho_3$, suggesting a natural order for grouping the eigenvectors in T as

$$T = [V(\omega_1), \dots, V(\omega_N), V(-\omega_1), \dots, V(-\omega_N)],$$

yielding

$$T^{-1}K_kT = \text{diag}(\omega_1, \dots, \omega_N, -\omega_1, \dots, -\omega_N), \\ T^{-1}M_kT = \text{diag}(\omega_1, \dots, \omega_N, \omega_1, \dots, \omega_N). \quad (\text{B12})$$

b. Determining the magnetic order

In the SB formalism, bosonic condensation occurs at specific momentum points $\pm\mathbf{k}_0$, corresponding to the minima of the spinon bands, which are obtained from the diagonalization of K_k . At a critical chemical potential μ_c , these points are associated with zero-energy eigenvectors Ψ_i . In real space, the condensate can then be expressed as a linear combination of these eigenvectors as (assuming for concreteness that $\mathbf{k}_0 \neq -\mathbf{k}_0$)

$$\langle \Psi \rangle = (\langle \hat{b}_{rA\uparrow} \rangle, \langle \hat{b}_{rB\uparrow} \rangle, \langle \hat{b}_{rA\downarrow}^\dagger \rangle, \langle \hat{b}_{rB\downarrow}^\dagger \rangle)^T \\ = e^{i\mathbf{k}_0 \cdot \mathbf{r}} (z_1 \Psi_1 + z_2 \Psi_2) \\ + e^{-i\mathbf{k}_0 \cdot \mathbf{r}} (z_3 \Psi_3 + z_4 \Psi_4) \quad \text{with } z_i \in \mathbb{C}. \quad (\text{B13})$$

The expectation value of the spins at positions $\mathbf{r}$ (spin texture) can then be obtained from

$$\langle S_\mu(\mathbf{r}) \rangle = X_\mu^\dagger \boldsymbol{\sigma} X_\mu, \quad (\text{B14})$$

up to global SR transformations, where $X_\mu = (\langle \hat{b}_{r\mu\uparrow} \rangle, \langle \hat{b}_{r\mu\downarrow} \rangle)^T$. Here μ refers to the sublattice indices and $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)^T$. Finally, spin-liquid *Ansätze* can be directly associated with classically magnetic ordered phases using these expressions. In what follows, we outline the key intermediate steps in employing this procedure for the *Ansätze* in Fig. 2 and the magnetic phases in Figs. 1(b) and 1(c).

(i) *The A_1 phase* U(1) $[0, 0, 0, \{0, 0, 0, 0\}, \{0, 0, 0, 0\}]$. The gauge-flux structure is given by only nonzero nearest-neighbor bonds $\mathcal{A}_{ij}$ (see Fig. 2):

$$\mathcal{A}_{12} = \mathcal{A}_1, \quad \mathcal{A}_{23} = \mathcal{A}_1, \quad \mathcal{A}_{34} = -\mathcal{A}_1, \quad \mathcal{A}_{41} = -\mathcal{A}_1, \\ \mathcal{A}_{13} = 0, \quad \mathcal{A}_{24} = 0, \\ \mathcal{B}_{12} = 0, \quad \mathcal{B}_{23} = 0, \quad \mathcal{B}_{34} = 0, \\ \mathcal{B}_{41} = 0, \quad \mathcal{B}_{13} = 0, \quad \mathcal{B}_{24} = 0. \quad (\text{B15})$$

For this state, we find fourfold degenerate spinon dispersions [related to M_k ; see (B12)]

$$\omega = \sqrt{\mathcal{A}_1^2 f_1(\mathbf{k}) + \mu^2} \\ \text{with } f_1(\mathbf{k}) = 2 \cos(k_x) \sin^2(k_y/2) + \cos(k_y). \quad (\text{B16})$$

The spinon condensation occurs at $\mathbf{Q} = (0, \pi)$ as the chemical potential approaches $\mu_c = 2\mathcal{A}_1$. There are four zero-energy eigenvectors, which are given by

$$\psi_1 = \frac{1}{N_{\psi_1}} \left(\frac{\mu_c - \sqrt{-4\mathcal{A}_1^2 + \mu_c^2}}{2\mathcal{A}_1}, 0, 0, 1 \right)^T, \\ \psi_2 = \frac{1}{N_{\psi_2}} \left(\frac{\mu_c + \sqrt{-4\mathcal{A}_1^2 + \mu_c^2}}{2\mathcal{A}_1}, 0, 0, 1 \right)^T, \\ \psi_3 = \frac{1}{N_{\psi_3}} \left(0, -\frac{\mu_c + \sqrt{-4\mathcal{A}_1^2 + \mu_c^2}}{2\mathcal{A}_1}, 1, 0 \right)^T, \\ \psi_4 = \frac{1}{N_{\psi_4}} \left(0, \frac{-\mu_c - \sqrt{-4\mathcal{A}_1^2 + \mu_c^2}}{2\mathcal{A}_1}, 1, 0 \right)^T, \quad (\text{B17})$$

where N_{ψ_i} represents the corresponding normalization factors for each eigenvector. For the cases where we do not analytically diagonalize K_k , condensation is reached numerically by introducing a small offset in $\mu \rightarrow \mu_c + \Delta\mu$, where $\Delta\mu \simeq 10^{-10}$. After considering equations (B13) and (B14), we calculate the observables $\mathcal{S}^{\text{EP}}$ and $\mathcal{X}^{\text{EP}}$ and conclude that the resulting magnetic order corresponds to the AM A_1 in Fig. 1(e).

(ii) *The A_2 phase* $Z_2 [0, \pi, 0, \{0, 0, 0, 0\}, \{0, 0, 0, 0\}]$. This *Ansatz* is obtained from the previous U(1) case by turning on the nearest-neighbor bonds $\mathcal{B}_{ij}$ as (see Fig. 2)

$$\mathcal{A}_{12} = \mathcal{A}_1, \quad \mathcal{A}_{23} = \mathcal{A}_1, \quad \mathcal{A}_{34} = -\mathcal{A}_1, \quad \mathcal{A}_{41} = -\mathcal{A}_1, \\ \mathcal{A}_{13} = 0, \quad \mathcal{A}_{24} = 0, \\ \mathcal{B}_{12} = i\mathcal{B}_1, \quad \mathcal{B}_{23} = i\mathcal{B}_1, \quad \mathcal{B}_{34} = -i\mathcal{B}_1, \quad \mathcal{B}_{41} = -i\mathcal{B}_1, \\ \mathcal{B}_{13} = 0, \quad \mathcal{B}_{24} = 0. \quad (\text{B18})$$

The corresponding dispersions are given by

$$\omega_{\pm} = \sqrt{(\mathcal{A}_1^2 - \mathcal{B}_1^2)f_2(\mathbf{k}) + \mu^2 \pm |\mathcal{B}_1 g(\mathbf{k})| \sqrt{\mathcal{A}_1^2 f_2(\mathbf{k}) + \mu^2}}, \quad (\text{B19})$$

with $f_2(\mathbf{k}) = 2 \cos(k_y) \sin^2(k_x/2) + \cos(k_x)$ and $g(\mathbf{k}) = 4 \sin(k_x/2) \cos(k_y/2)$. The spinon condensation takes place at the point $\mathbf{Q} = (\pi, 0)$ as the chemical potential approaches $\mu_c = -2\sqrt{\mathcal{A}_1^2 + \mathcal{B}_1^2}$. The associated zero-energy eigenvectors to the condensation are expressed as

$$\psi_1 = \frac{1}{N_{\psi}} \left(-\frac{\sqrt{\mathcal{A}_1^2 + \mathcal{B}_1^2}}{\mathcal{A}_1}, i\frac{\mathcal{B}_1}{\mathcal{A}_1}, 0, 1 \right)^T, \\ \psi_2 = \frac{1}{N_{\psi}} \left(i\frac{\mathcal{B}_1}{\mathcal{A}_1}, \frac{\sqrt{\mathcal{A}_1^2 + \mathcal{B}_1^2}}{\mathcal{A}_1}, 1, 0 \right)^T, \quad (\text{B20})$$

with N_{ψ} representing the normalization factor. The resulting magnetic order associated with the condensation is the canted AM A_2 phase in Fig. 1(e).

(iii) *The B_1 phase* $U(1)$ $[0, 0, 0, \{0, 0, 0, 0\}, \{0, 0, 0, 0\}]$. The gauge-flux structure is given by real nearest-neighbor and next-nearest-neighbor bonds $\mathcal{A}_{ij}$ (see Fig. 2):

$$\begin{aligned} \mathcal{A}_{12} &= \mathcal{A}_1, & \mathcal{A}_{23} &= 0, & \mathcal{A}_{34} &= -\mathcal{A}_1, & \mathcal{A}_{41} &= 0, \\ \mathcal{A}_{13} &= \mathcal{A}_2, & \mathcal{A}_{24} &= -\mathcal{A}_2, \end{aligned}$$

$$\begin{aligned} \mathcal{B}_{12} &= 0, & \mathcal{B}_{23} &= 0, & \mathcal{B}_{34} &= 0, & \mathcal{B}_{41} &= 0, \\ \mathcal{B}_{13} &= 0, & \mathcal{B}_{24} &= 0. \end{aligned} \quad (\text{B21})$$

This leads to doubly degenerate bands whose dispersions are

$$\omega_{\pm} = \frac{1}{2} \sqrt{2\mathcal{A}_1^2 f_3(\mathbf{k}) + 4\mu^2 - 2\mathcal{A}_2^2 f_4(\mathbf{k}) \pm |\mathcal{A}_2 m_+(\mathbf{k})| \sqrt{-8\mathcal{A}_1^2 f_3(\mathbf{k}) - 2\mathcal{A}_2^2 m_+^2(\mathbf{k})}}, \quad (\text{B22})$$

with $f_3(\mathbf{k}) = \cos(k_x + k_y) - 1$, $f_4(\mathbf{k}) = \sin^2(k_x) + \sin^2(k_y)$, and $m_{\pm}(\mathbf{k}) = \sin(k_x) \pm \sin(k_y)$. Bosonic condensation takes place at $\mu_c = \mathcal{A}_1 + \mathcal{A}_2$ and $\mathbf{Q} = \pm(\pi/2, \pi/2)$. Approaching the condensation numerically with $\mathcal{A}_1 = -0.2$ and $\mathcal{A}_2 = -0.5$, the zero-energy eigenvectors at $\mathbf{Q}$ are given by $\psi_1 = \frac{1}{2}(1, 1, i, i)^T$ and $\psi_2 = \frac{1}{2}(-i, -i, 1, 1)^T$, with B_1 the associated magnetic order.

(iv) *The B_2 phase* $U(1)$ $[0, 0, 0, \{0, 0, 0, 0\}, \{0, \pi, 0, 0\}]$. The gauge-flux structure is given by both real nearest-neighbor and next-nearest-neighbor bonds $\mathcal{A}_{ij}$ and $\mathcal{B}_{ij}$ (see Fig. 2):

$$\begin{aligned} \mathcal{A}_{12} &= 0, & \mathcal{A}_{23} &= 0, & \mathcal{A}_{34} &= -\mathcal{A}_1, & \mathcal{A}_{41} &= -\mathcal{A}_1, \\ \mathcal{A}_{13} &= 0, & \mathcal{A}_{24} &= \mathcal{A}_2, & \mathcal{B}_{12} &= \mathcal{B}_1, & \mathcal{B}_{23} &= \mathcal{B}_1, \\ \mathcal{B}_{34} &= 0, & \mathcal{B}_{41} &= 0, & \mathcal{B}_{13} &= \mathcal{B}_2, & \mathcal{B}_{24} &= 0. \end{aligned} \quad (\text{B23})$$

It leads to doubly degenerate bands. The spinon condensation takes place at $\mathbf{Q} = \pm(\pi/2, 0)$ as we approach the

critical chemical potential given by $\mu_c = \frac{1}{2}(\mathcal{A}_2 + \mathcal{B}_2) - \frac{1}{2}\sqrt{\mathcal{A}_2^2 + 4\mathcal{B}_1^2 - 2\mathcal{A}_2\mathcal{B}_2 + \mathcal{B}_2^2}$. Approaching the condensation numerically with $\mathcal{A}_1 = -0.12$, $\mathcal{A}_2 = -0.35$, $\mathcal{B}_1 = -0.6$, and $\mathcal{B}_2 = 0.3$, we find the eigenvectors at $\mathbf{Q}$ to be given by $\psi_1 = (0.60752, 0.36185, 0.60750i, 0. + 0.36183i)^T$ and $\psi_2 = (-0.6075i, -0.36183i, 0.60752, 0.36185)^T$. The resulting magnetic order associated with the condensation is the B_2 phase in Fig. 1(c).

(v) *The A_3 phase* $Z_2[\pi, 0, 0, \{0, 0, 0, 0\}, \{0, 0, 0, 0\}]$. The gauge-flux structure is given by real nearest-neighbor and next-nearest-neighbor bonds $\mathcal{A}_{ij}$ (see Fig. 2):

$$\begin{aligned} \mathcal{A}_{12} &= -\mathcal{A}_1, & \mathcal{A}_{23} &= \mathcal{A}_1, & \mathcal{A}_{34} &= -\mathcal{A}_1, & \mathcal{A}_{41} &= -\mathcal{A}_1, \\ \mathcal{A}_{13} &= -\mathcal{A}_2, & \mathcal{A}_{24} &= -\mathcal{A}_2, & \mathcal{B}_{12} &= 0, & \mathcal{B}_{23} &= 0, \\ \mathcal{B}_{34} &= 0, & \mathcal{B}_{41} &= 0, & \mathcal{B}_{13} &= 0, & \mathcal{B}_{24} &= 0. \end{aligned} \quad (\text{B24})$$

In this case, we discover the spinon bands to be doubly degenerate bands, given by

$$\omega_{\pm} = \frac{1}{2} \sqrt{4\mathcal{A}_1^2 f_5(\mathbf{k}) + 4\mu^2 - 2\mathcal{A}_2^2 f_4(\mathbf{k}) \pm \sqrt{2}|\mathcal{A}_2 m_-(\mathbf{k})| \sqrt{-8\mathcal{A}_1^2 f_5(\mathbf{k}) + 2\mathcal{A}_2^2 m_+^2(\mathbf{k})}}, \quad (\text{B25})$$

with $f_5(\mathbf{k}) = \sin(k_x) \sin(k_y) - 1$, $f_4(\mathbf{k}) = \sin^2(k_x) + \sin^2(k_y)$, and $m_{\pm}(\mathbf{k}) = \sin(k_x) \pm \sin(k_y)$. The minimum of the dispersion appears at $\pm\mathbf{Q} = \pm(-\pi/2, \pi/2)$, with critical chemical potential given by $\mu_c = -|\sqrt{2}\mathcal{A}_1 + \mathcal{A}_2|$. For $\mathcal{A}_1 = -0.3$ and $\mathcal{A}_2 = -0.5$, the normalized eigenvectors are given by $\psi_1 = (0.353553(1-i), -0.5i, 0.353554(1+i), 0.5)^T$ and $\psi_2 = (0.5, 0.353554(1-i), 0.5i, 0.353553(1+i))^T$. The resulting magnetic order associated with the condensation is the A_3 phase in Fig. 1(c).

(vi) *The C phase* $Z_2[\pi, \pi, \pi, \{\frac{3\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{2}\}, \{\frac{3\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{2}\}]$. Finally, the *Ansatz* related to the orbital AM C takes into account all types of bonds $\mathcal{A}_{ij}$ and $\mathcal{B}_{ij}$, i.e.,

$$\begin{aligned} \mathcal{A}_{12} &= -\mathcal{A}_1, & \mathcal{A}_{23} &= \mathcal{A}_1, & \mathcal{A}_{34} &= -\mathcal{A}_1, & \mathcal{A}_{41} &= -\mathcal{A}_1, \\ \mathcal{A}_{13} &= -\mathcal{A}_2, & \mathcal{A}_{24} &= -\mathcal{A}_2, & \mathcal{B}_{12} &= i\mathcal{B}_1, & \mathcal{B}_{23} &= i\mathcal{B}_1, \\ \mathcal{B}_{34} &= i\mathcal{B}_1, & \mathcal{B}_{41} &= -i\mathcal{B}_1, & \mathcal{B}_{13} &= i\mathcal{B}_2, & \mathcal{B}_{24} &= -i\mathcal{B}_2. \end{aligned} \quad (\text{B26})$$

This results in two doubly degenerate bands which display minima at $\mathbf{Q} = \pm(-\pi/2, \pi/2)$. For $\mathcal{A}_1 = -0.3$, $\mathcal{A}_2 = -0.2$, $\mathcal{B}_1 = -0.35$, and $\mathcal{B}_2 = -0.1$, the zero-energy eigenvectors are given by $\psi_1 = (0.41574 + 0.17646i, 0.54409, 0.38472 + 0.38472i, -0.41874 + 0.16920i)^T$ and $\psi_2 = (0.41874 -$

$0.16919i, 0.38472 - 0.38472i, 0.54409, -0.17646 + 0.41574i)^T$.

In addition, the spin correlations $\mathcal{S}^{\text{EP}}$ and the spin scalar chiralities χ^{EP} can also be calculated in the spin-liquid regime to be further compared with the same observables for the classically magnetic orders. In the quantum spin-liquid (QSL) regime, the expectation values are determined by evaluating all possible Wick contractions of the Schwinger boson operators in $\hat{S}_i \cdot \hat{S}_j$ and $\hat{S}_i \cdot (\hat{S}_j \times \hat{S}_k)$. Using Eqs. (B1) and (B3), these are given by

$$\langle \hat{S}_i \cdot \hat{S}_j \rangle = \frac{3}{2} \left(\frac{|\mathcal{B}_{ij}|^2 - |\mathcal{A}_{ij}|^2}{J_{ij}^2} \right) \quad (\text{B27})$$

and

$$\begin{aligned} \langle \hat{S}_i \cdot (\hat{S}_j \times \hat{S}_k) \rangle &= -\frac{3}{J_{ij}J_{jk}J_{ki}} [\text{Im}(\mathcal{B}_{ij}\mathcal{B}_{jk}\mathcal{B}_{ki}) \\ &\quad + \text{Im}(\mathcal{A}_{ij}\mathcal{A}_{jk}^*\mathcal{B}_{ki}) + \text{Im}(\mathcal{B}_{ij}\mathcal{A}_{jk}\mathcal{A}_{ki}^*) \\ &\quad + \text{Im}(\mathcal{A}_{ij}^*\mathcal{B}_{jk}\mathcal{A}_{ki})]. \end{aligned} \quad (\text{B28})$$

These quantities in the QSL regime are gauge invariant, as expected. Moreover, Eq. (B28) confirms that a mean-field *Ansatz* based on purely real $\mathcal{A}_{ij}$ and $\mathcal{B}_{ij}$ is

TABLE II. Values of the observables $\mathcal{S}^{\text{EP}}$ and χ^{EP} according to Eqs. (B27) and (B28) before (in the quantum spin liquid regime) and after bosonic condensation (for the equivalent classical magnetic order). The latter is calculated using Eqs. (B13) and (B14). For the QSLs, the observables are presented up to a multiplicative factor. The nonuniversal parameters $s_0, s', s_1, \chi_1 \in \mathbb{R}^+$ depend on the mean-field values $\mathcal{A}_{ij}$ and $\mathcal{B}_{ij}$.

Label	QSL		Condensate	
	$\mathcal{S}^{\text{EP}}$	χ^{EP}	$\mathcal{S}^{\text{EP}}$	χ^{EP}
A_1	$(-1, -1, -1, -1)$	$(0,0,0,0)$	$(-1, -1, -1, -1)$	$(0,0,0,0)$
A_2	$(-1, -1, -1, -1)$	$(0,0,0,0)$	$s_0(-1, -1, -1, -1)$	$(0,0,0,0)$
A_3	$(-1, -1, -1, -1)$	$(0,0,0,0)$	$(0,0,0,0)$	$(0,0,0,0)$
B_1	$(-1, 0, -1, 0)$	$(0,0,0,0)$	$(-1, 1, -1, 1)$	$(0,0,0,0)$
B_2	$(1, 1, -s_1, -s_1)$	$(0,0,0,0)$	$(1, 1, -1, -1)$	$(0,0,0,0)$
C	$(-1, -1, -1, -1)$	$(1, -1, 1, -1)$	$s'(-1, -1, -1, -1)$	$\chi_1(-1, 1, -1, 1)$

incapable of generating spin liquids with nonzero χ^{EP} and, by extension, noncoplanar magnetic phases upon bosonic condensation. More specifically, by using the *Ansatz* $Z_2[\pi, \pi, \pi, \{\frac{3\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{2}\}, \{\frac{3\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{2}\}]$ we confirm that the expected staggered chirality pattern for phase C can be found in both regimes (see Table II).

2. $\mathbb{CP}^1$ action for A_1

In this Appendix we present the continuum-field-theoretic description of the transition from the spin-liquid phase to altermagnetic order in the case of the A_1 phase. Following Read and Sachdev's procedure [81,104], we parametrize the bosonic operators in A and B sublattices using slowly varying fields, i.e.,

$$b_{A,r_i,\alpha} = \psi_{A,\alpha}(\mathbf{r}_i) e^{i\mathbf{Q} \cdot \mathbf{r}_i}, \quad (\text{B29})$$

$$b_{B,r_i,\alpha} = \sigma_{\alpha\beta}^y \psi_{B,\beta}(\mathbf{r}_i) e^{i\mathbf{Q} \cdot \mathbf{r}_i}, \quad (\text{B30})$$

where $\mathbf{Q}$ is the position in momentum space where spinon condensation takes place. With $\mathcal{A}_1 < 0$, in the case of our *Ansatz* for the A_1 phase (as described in Fig. 2 of the main text), we found dispersion minimum to be at $\mathbf{Q} = (0, \pi)$. To derive the continuum field theory, we plug Eqs. (B29) and (B30) into the mean-field Hamiltonian (B4) and carry out a gradient expansion, which leads to (in an action description)

$$S = \int \frac{d^2\mathbf{r}}{a^2} \int d\tau \left[\left(\psi_{A,\alpha}^* \frac{d}{d\tau} \psi_{A,\alpha} + \psi_{B,\alpha}^* \frac{d}{d\tau} \psi_{B,\alpha} \right) - \mu(\psi_{A,\alpha}^* \psi_{A,\alpha} + \psi_{B,\alpha}^* \psi_{B,\alpha}) + 2\mathcal{A}_1(\psi_{A,\alpha} \psi_{B,\alpha} + \psi_{A,\alpha}^* \psi_{B,\alpha}^*) - \frac{\mathcal{A}_1 a^2}{4} (\nabla \psi_{A,\alpha} \nabla \psi_{B,\alpha} + \nabla \psi_{A,\alpha}^* \nabla \psi_{B,\alpha}^*) \right]. \quad (\text{B31})$$

Next we introduce two fields defined by $z_\alpha = \frac{\psi_{A,\alpha} + \psi_{B,\alpha}^*}{\sqrt{2}}$ and $\pi_\alpha = \frac{\psi_{A,\alpha} - \psi_{B,\alpha}^*}{\sqrt{2}}$ and rewrite the action as

$$S = \int \frac{d^2\mathbf{r}}{a^2} \int d\tau \left[\left(z_\alpha^* \frac{d}{d\tau} \pi_\alpha + \pi_\alpha^* \frac{d}{d\tau} z_\alpha \right) + (2\mathcal{A}_1 - \mu)|z_\alpha|^2 - (2\mathcal{A}_1 + \mu)|\pi_\alpha|^2 - \frac{\mathcal{A}_1 a^2}{4} (|\nabla z_\alpha|^2 - |\nabla \pi_\alpha|^2) \right]. \quad (\text{B32})$$

As can be seen from Eq. (B32), in this new form, π_α fields turn out to be massive, whereas z_α fields become massless near the critical $\mu_c = 2\mathcal{A}_1$, marking the phase transition from the $U(1)[0, 0, 0, \{0, 0, 0, 0\}, \{0, 0, 0, 0\}]$ spin liquid to altermagnetic order A_1 . Note that the critical chemical potential μ_c is the same as the one obtained previously within the SB framework, exemplifying the consistency between both approaches. Finally, integrating the π_α fields and restoring gauge invariance by introduction of the gauge field a_μ , we arrive at the effective action

$$S_{\text{eff}} = \int \frac{d^2\mathbf{r}}{4a} \int_0^{c\beta} d\bar{\tau} \left(|(\partial_\mu - ia_\mu)z_\alpha|^2 + \frac{\Delta^2}{c^2} |z_\alpha|^2 \right). \quad (\text{B33})$$

This essentially describes the massive spinons z_α coupled to a compact $U(1)$ gauge field, where $c = |\mathcal{A}_1|a$ is the spinon velocity, $\Delta^2 = (\mu^2 - 4\mathcal{A}_1^2)$, and $a_\tau = \frac{a}{c}$.

APPENDIX C: ELECTRONIC SPECTRAL FUNCTION

In this Appendix we explain the details on how the electronic properties inside the fractionalized AM phases in Fig. 3 of the main text are computed. The total spin-fermion-model-like action $\mathcal{S} = \mathcal{S}_e + \mathcal{S}_c + \mathcal{S}_\Phi$ we start from consists of three parts. First is the noninteraction electronic action

$$\mathcal{S}_e = \int_0^\beta d\tau \left(\sum_{\mathbf{R},\ell} c_{\mathbf{R},\ell,\sigma}^\dagger (\partial_\tau - \mu) c_{\mathbf{R},\ell,\sigma} - \sum_{\mathbf{R},\mathbf{R}',\ell,\ell'} t_{\mathbf{R},\ell;\mathbf{R}',\ell'} c_{\mathbf{R},\ell,\sigma}^\dagger c_{\mathbf{R}',\ell',\sigma} \right), \quad (\text{C1})$$

where $c_{\mathbf{R},\ell,\sigma}^\dagger$ are the (τ -dependent) electronic fields associated with the creation of an electron in the unit cell labeled by the Bravais-lattice site $\mathbf{R}$, of sublattice $\ell = A, B$, and of spin σ . Note that $\mathcal{S}_e$ is invariant under global spin rotation (summation over repeated σ indices is implied). Although most of the following analysis is more generally valid, we later choose the hopping matrix elements $t_{\mathbf{R},\ell;\mathbf{R}',\ell'}$ as described in the main text with nearest- and next-nearest-neighbor hopping on the lattice in Fig. 1(a). These electrons are coupling via $[\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)^T]$ is a vector of Pauli matrices]

$$\mathcal{S}_c = \int_0^\beta d\tau \sum_{\mathbf{R},\ell} c_{\mathbf{R},\ell,\sigma}^\dagger \boldsymbol{\sigma}_{\sigma,\sigma'} c_{\mathbf{R},\ell,\sigma'} \cdot \boldsymbol{\Phi}_{\mathbf{R},\ell} \quad (\text{C2})$$

to a bosonic field Φ which describes fluctuations of the electronic spins. While at high energies the action S_Φ of Φ can just be obtained via a Hubbard-Stratonovich decoupling of some bare electronic interactions, here we think of S as an effective low-energy action, just as in the celebrated spin-fermion model [105], and take

$$S_\Phi = \frac{1}{4g_0} \int_0^\beta d\tau \left(\sum_{\mathbf{R},\ell} [(\partial_\tau \Phi_{\mathbf{R},\ell})^2 + V(\Phi_{\mathbf{R},\ell}^2)] + \sum_{\mathbf{R},\mathbf{R}',\ell,\ell'} \mathcal{J}_{\mathbf{R},\ell;\mathbf{R}',\ell'} \Phi_{\mathbf{R},\ell} \cdot \Phi_{\mathbf{R}',\ell'} \right). \quad (\text{C3})$$

Here $\mathcal{J}_{\mathbf{R},\ell;\mathbf{R}',\ell'}$ are effective exchange coupling constants, akin to J_j in the spin model (1).

To describe the fractionalized phases, characterized by topological order, we transform into a rotating reference frame in spin space via [84–86]

$$c_{\mathbf{R},\ell,\sigma}(\tau) = \sum_{\alpha=\pm} [R_{\mathbf{R},\ell}(\tau)]_{\sigma,\alpha} \psi_{\mathbf{R},\ell,\alpha}(\tau). \quad (\text{C4})$$

In Eq. (C4), R are (space-time-dependent, bosonic) SU(2) matrices, physically related to the spinons and thus to the Schwinger bosons in Eq. (B1) in the spin-liquid regime. While the spin is carried by R , the charge is governed by the fermionic chargin fields ψ ; to indicate that the latter do not carry spin, we use $\alpha = \pm$ (rather than $\sigma = \uparrow, \downarrow$) to label its two components. As the physical electrons are invariant under the local reparametrization

$$R_{\mathbf{R},\ell}(\tau) \rightarrow R_{\mathbf{R},\ell}(\tau) V_{\mathbf{R},\ell}^\dagger(\tau), \\ \psi_{\mathbf{R},\ell}(\tau) \rightarrow V_{\mathbf{R},\ell}(\tau) \psi_{\mathbf{R},\ell}(\tau), \quad V_{\mathbf{R},\ell}^\dagger(\tau) V_{\mathbf{R},\ell}(\tau) = \mathbb{1}, \quad (\text{C5})$$

inserting Eq. (C4) into S , as we do in the following, will lead to an SU(2) gauge theory.

To discuss the action of spinons and chargons, we start with the coupling S_c , which can be exactly rewritten as

$$S_c = \int_0^\beta d\tau \sum_{\mathbf{R},\ell} \psi_{\mathbf{R},\ell,\alpha}^\dagger \sigma_{\alpha,\alpha'} \psi_{\mathbf{R},\ell,\alpha'} \cdot \mathbf{H}_{\mathbf{R},\ell}, \quad (\text{C6})$$

where $\mathbf{H}$ plays the role of a Higgs field, transforming under the adjoint representation of the SU(2) gauge transformation (C5). It is formally related to the collective boson Φ via

$$\mathbf{H}_{\mathbf{R},\ell} \cdot \sigma = \Phi_{\mathbf{R},\ell} \cdot R_{\mathbf{R},\ell}^\dagger \sigma R_{\mathbf{R},\ell} \quad \text{and thus } (\Phi_{\mathbf{R},\ell})_a \\ = \frac{1}{2} \text{tr}(\sigma_a R_{\mathbf{R},\ell} \sigma R_{\mathbf{R},\ell}^\dagger) \cdot \mathbf{H}_{\mathbf{R},\ell}. \quad (\text{C7})$$

Inserting the parametrization (C4) into the noninteracting electronic part S_e of the action will lead to quartic terms; as in Ref. [90], we will decouple those into chargin and spinon parts. This leads to the chargin action

$$S_\psi = \int_0^\beta d\tau \left(\sum_{\mathbf{R},\ell} \psi_{\mathbf{R},\ell,\alpha}^\dagger (\partial_\tau - \mu) \psi_{\mathbf{R},\ell,\alpha} - \sum_{\mathbf{R},\mathbf{R}',\ell,\ell'} t_{\mathbf{R},\ell;\mathbf{R}',\ell'} \psi_{\mathbf{R},\ell,\alpha}^\dagger (\mathcal{U}_{\mathbf{R},\ell;\mathbf{R}',\ell'})_{\alpha,\alpha'} \psi_{\mathbf{R}',\ell',\alpha'} \right), \quad (\text{C8})$$

where $(\mathcal{U}_{\mathbf{R},\ell;\mathbf{R}',\ell'})_{\alpha,\alpha'} = \langle (R_{\mathbf{R},\ell}^\dagger R_{\mathbf{R}',\ell'})_{\alpha,\alpha'} \rangle$ and the spinon contribution

$$S_R^e = \int_0^\beta d\tau \text{tr} \left(\sum_{\mathbf{R},\ell} \chi_{\mathbf{R},\ell;\mathbf{R},\ell}^T R_{\mathbf{R},\ell}^\dagger \partial_\tau R_{\mathbf{R},\ell} - \sum_{\mathbf{R},\mathbf{R}',\ell,\ell'} t_{\mathbf{R},\ell;\mathbf{R}',\ell'} \chi_{\mathbf{R},\ell;\mathbf{R}',\ell'}^T R_{\mathbf{R},\ell}^\dagger R_{\mathbf{R}',\ell'} \right), \quad (\text{C9})$$

with the 2×2 matrices $(\chi_{\mathbf{R},\ell;\mathbf{R}',\ell'})_{\alpha,\alpha'} = \langle \psi_{\mathbf{R},\ell,\alpha}^\dagger \psi_{\mathbf{R}',\ell',\alpha'} \rangle$.

Finally, plugging Eq. (C7) into S_Φ in Eq. (C3) leads to coupling terms between the Higgs field and the spinons. To describe the fractionalized phase with topological order, we follow previous works (see, e.g., Ref. [90]) and keep the Higgs field at the saddle-point value $\mathbf{H}_{\mathbf{R},\ell} \rightarrow \langle \mathbf{H}_{\mathbf{R},\ell} \rangle$. Note that $\langle \mathbf{H}_{\mathbf{R},\ell} \rangle \neq 0$ does *not* break SR symmetry as the Higgs field is invariant under spin rotations [cf. Eq. (C6)], as long as the bosons R remain gapped (not condensed). We can now associate nonmagnetic fractionalized states with each of the phases in the classical phase diagram in Fig. 1(b) by choosing $\langle \mathbf{H}_{\mathbf{R},\ell} \rangle$ to be equal to the respective spin texture in the gauge where the bosons $R_{\mathbf{R},\ell}$ will condense in a spatially uniform way, $\langle R_{\mathbf{R},\ell} \rangle = R_0$, once the gap closes. In that case, we see in Eq. (C7) that $\langle \Phi_{\mathbf{R},\ell} \rangle$ will be equal to $\langle \mathbf{H}_{\mathbf{R},\ell} \rangle$ apart from a global spin rotation. To keep the discussion concise, we focus on phase A₁, the collinear AM, in the following; without loss of generality we choose a gauge where

$$\langle \mathbf{H}_{\mathbf{R},\ell} \rangle = H_0 \hat{e}_z s_\ell, \quad \text{with } s_A = -s_B = 1. \quad (\text{C10})$$

In this gauge, it is convenient to use the $\mathbb{CP}^1$ parametrization

$$R_{\mathbf{R},\ell} = \begin{pmatrix} (z_{\mathbf{R},\ell})_\uparrow & -(z_{\mathbf{R},\ell}^*)_\downarrow \\ (z_{\mathbf{R},\ell})_\downarrow & (z_{\mathbf{R},\ell}^*)_\uparrow \end{pmatrix}, \quad \text{with } z_{\mathbf{R},\ell}^\dagger z_{\mathbf{R},\ell} = 1. \quad (\text{C11})$$

The reason is that this leads to the intuitive form $N_{\mathbf{R},\ell} = z_{\mathbf{R},\ell}^\dagger \sigma z_{\mathbf{R},\ell}$ of the normalized altermagnetic order parameter $N_{\mathbf{R},\ell} = s_\ell \Phi_{\mathbf{R},\ell} / |\Phi_{\mathbf{R},\ell}|$, as readily follows from Eq. (C7). We next express S_Φ in Eq. (C3) in terms of $N_{\mathbf{R},\ell}$, yielding

$$S_\Phi = \frac{1}{4g} \int_0^\beta d\tau \left(\sum_{\mathbf{R},\ell} (\partial_\tau N_{\mathbf{R},\ell})^2 + \sum_{\mathbf{R},\mathbf{R}',\ell,\ell'} s_\ell s_{\ell'} \mathcal{J}_{\mathbf{R},\ell;\mathbf{R}',\ell'} N_{\mathbf{R},\ell} \cdot N_{\mathbf{R}',\ell'} \right), \quad g = g_0/H_0^2, \quad (\text{C12})$$

which we next study in the continuum limit; to this end, we introduce the continuum fields $\mathbf{n}(\tau, \mathbf{x})$ and $\mathbf{m}(\tau, \mathbf{x})$ with the constraints $\mathbf{n}^2 = 1$ and $\mathbf{n} \cdot \mathbf{m} = 0$ and write

$$N_{\mathbf{R},\ell}(\tau) = \mathbf{n}(\tau, \mathbf{R} + \mathbf{v}_\ell) \sqrt{1 - \mathbf{m}^2(\tau, \mathbf{R} + \mathbf{v}_\ell)} \\ + s_\ell \mathbf{m}(\tau, \mathbf{R} + \mathbf{v}_\ell), \\ \text{with } \mathbf{v}_A = \hat{\mathbf{x}}/2, \quad \mathbf{v}_B = \hat{\mathbf{y}}/2. \quad (\text{C13})$$

Including nearest- (J_1) and next-nearest-neighbor (J_2) exchange couplings, just like in the spin model we study [see Fig. 1(a)], and expanding up to second order in gradients and

$\mathbf{m}$, we find the nonlinear sigma model

$$S_\Phi = \frac{1}{4g} \int_0^\beta d\tau d^2\mathbf{x} [(\partial_\tau \mathbf{n})^2 + v^2(\partial_x \mathbf{n})^2 + v^2(\partial_y \mathbf{n})^2 + r\mathbf{m}^2], \quad (\text{C14})$$

with $v^2 = J_1/2 - J_2/4$ and $r = 8J_1$. As expected, the ferromagnetic fluctuations $\mathbf{m}$ are massive, $r > 0$, and can thus be neglected, i.e., $N_{\mathbf{R},\ell}(\tau) \simeq \mathbf{n}(\tau, \mathbf{R} + \mathbf{v}_\ell)$. The AM fluctuations $\mathbf{n}$ are governed by the usual relativistic form in Eq. (C14), which in turn can be restated [88,106] as the $\mathbb{CP}^1$ model

$$S_\Phi^{z,c} = g^{-1} \int d\tau d^2\mathbf{x} |(\partial_\mu - ia_\mu)z_\alpha|^2, \quad (\text{C15})$$

where a_μ is an emergent U(1) gauge field; this is exactly the same continuum field theory we derived for the SB mean-field Ansatz for phase A₁ in Appendix B 2. As in Ref. [90], we will neglect the gauge-field fluctuations, which are not expected to change the results qualitatively [107], and work on the lattice, i.e., use

$$S_\Phi^{z,c} = g^{-1} T \sum_{\Omega_n} \sum_q z_{n,q}^\dagger z_{n,q} (\Omega_n^2 + \mathcal{E}_q^2),$$

$$\mathcal{E}_q^2 = 2J^2(2 - \cos q_x - \cos q_y) + \Delta^2 \quad (\text{C16})$$

for the bosons in Eq. (C11); here Δ is the bosonic gap that has to be finite in the spin-liquid regime with topological order and is determined by the condition $\langle z_{\mathbf{R},\ell}^\dagger z_{\mathbf{R},\ell} \rangle = 1$. For the numerics presented in Figs. 3(d)–3(f) of the main text, we used $J/t_1 = 1$ and $\Delta/t_1 = 0.01$.

With the spinon action (C16) at hand, we can simplify the form of the chargon action in Eq. (C8). In the gauge leading to Eq. (C16), we immediately conclude that $\mathcal{U}_{\mathbf{R},\ell;\mathbf{R}',\ell'} = \text{diag}(\langle z_{\mathbf{R},\ell}^\dagger z_{\mathbf{R}',\ell'} \rangle, \langle z_{\mathbf{R}',\ell'}^\dagger z_{\mathbf{R},\ell} \rangle)$. As the spinon dispersion $\mathcal{E}_q$ is even in $\mathbf{q}$, the two terms on the diagonal of $\mathcal{U}_{\mathbf{R},\ell;\mathbf{R}',\ell'}$ are identical (and real) such that we are left with

$$\mathcal{U}_{\mathbf{R},\ell;\mathbf{R}',\ell'} = \mathbb{1} Z_{\mathbf{R},\ell;\mathbf{R}',\ell'}, \quad Z_{\mathbf{R},\ell;\mathbf{R}',\ell'} \in \mathbb{R}. \quad (\text{C17})$$

As the spinon (and chargon) theory is further invariant under all spatial symmetries, there will only be two independent values Z and Z' , renormalizing the nearest- (t) and next-nearest-neighbor hopping (t') of the chargons relative to the

bare electrons $t \rightarrow Zt$ and $t' \rightarrow Z't'$, respectively. One can determine Z and Z' by solving the chargon and spinon theory self-consistently [90], yielding order-1 values. To facilitate the comparison of the noninteracting spectra and the spectral function in the fractionalized theory in Fig. 3, we simply set $Z = Z' = 1$ in the following. Including the contribution from the Higgs field, the full chargon action can be written as

$$S_\psi + S_c = T \sum_{\omega_n} \frac{1}{N} \sum_k \psi_{k,\alpha}^\dagger (-i\omega_n + \epsilon_k \tau_0 + \mathbf{g}_k^\alpha \cdot \boldsymbol{\tau}) \psi_{k,\alpha}, \quad (\text{C18})$$

where τ_j are Pauli matrices in sublattice space and

$$\epsilon_k = -t'(\cos k_x + \cos k_y) - \mu, \quad (\text{C19})$$

$$(\mathbf{g}_k^\alpha)_x + i(\mathbf{g}_k^\alpha)_y = t(1 + e^{ik_x} + e^{i(k_x - k_y)} + e^{-ik_y}),$$

$$(\mathbf{g}_k^\alpha)_z = -t'(\cos k_x - \cos k_y) + H_0\alpha. \quad (\text{C20})$$

Consequently, the chargon Green's function (a matrix in sublattice space) reads

$$G_{\alpha,\alpha'}^\psi(i\omega_n, \mathbf{k}) = \delta_{\alpha,\alpha'} \frac{i\omega_n - \epsilon_k + \mathbf{g}_k^\alpha \cdot \boldsymbol{\tau}}{(i\omega_n - E_{k,\alpha}^+)(i\omega_n - E_{k,\alpha}^-)},$$

$$E_{k,\alpha}^\pm = \epsilon_k \pm |\mathbf{g}_k^\alpha|. \quad (\text{C21})$$

Finally, we can compute the electron Green's function $G_e(i\omega_n, \mathbf{k})$, which by virtue of rewriting (C4) becomes a convolution of the chargon and spinon Green's function [see also the diagram in Fig. 3(c)]. After straightforward algebra, we find

$$[G_e(i\omega_n, \mathbf{k})]_{\ell,\sigma;\ell',\sigma'} = \delta_{\sigma,\sigma'} T \sum_{\Omega_m} \frac{1}{N} \sum_q \sum_{\alpha=\pm} G_z(i\Omega_m, \mathbf{q}) G_{\alpha,\alpha}^\psi \times (i\omega_n - i\Omega_m, \mathbf{k} - \mathbf{q}) e^{iq(\mathbf{v}_{\ell'} - \mathbf{v}_\ell)}, \quad (\text{C22})$$

where $G_z(i\Omega_n, \mathbf{q}) = g/(\Omega_n^2 + \mathcal{E}_q^2)$ is the spinon Green's function associated with Eq. (C16). We can directly see that the convolution with the spinons restored full SU(2) SR invariance (proportional to $\delta_{\sigma,\sigma'}$). More explicitly, the resultant spectral function we plot in the main text is written as

$$\mathcal{A}_k(\omega) = -\frac{1}{4\pi} \text{Im tr}[G_e(\omega + i\eta, \mathbf{k})] \quad (\text{C23})$$

$$= -\frac{1}{\pi N} \text{Im} \sum_q \left(T \sum_{\Omega_m} \sum_{\alpha=\pm} \frac{g}{\Omega_m^2 + \mathcal{E}_q^2} \frac{i\omega_n - i\Omega_m - \epsilon_{\mathbf{k}-\mathbf{q}} + \mathbf{g}_{\mathbf{k}-\mathbf{q}}^\alpha \cdot \boldsymbol{\tau}}{(i\omega_n - i\Omega_m - E_{\mathbf{k}-\mathbf{q},\alpha}^+)(i\omega_n - i\Omega_m - E_{\mathbf{k}-\mathbf{q},\alpha}^-)} \right)_{i\omega_n \rightarrow \omega + i\eta}. \quad (\text{C24})$$

As the spinon gap Δ has a more direct physical meaning than the coupling constant g , we fix Δ and then think of g as a function of Δ (and not vice versa), $g = g(\Delta)$. It was shown in Ref. [90] that the above approximations leave the frequency sum rule

$$\int_{-\infty}^{\infty} d\omega A_k(\omega) = 1 \quad (\text{C25})$$

untouched. We use Eq. (C25) to determine g for each Δ , yielding the nontrivial check that a *single* g can ensure that Eq. (C25) holds at *all* $\mathbf{k}$.

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